$$m_{ji} = m_{ij}^*.$$

In other words, if you flip a Hermitian matrix about the main diagonal and then take its complex conjugate, the result is the same as the original matrix. Hermitian operators (and matrices) have some special properties. The first is that their eigenvalues are all real. Let's prove it.

Suppose λ and |λ> represent an eigenvalue and the corresponding eigenvector of the Hermitian operator **L**. In symbols,

$$\mathbf{L}|\lambda\rangle = \lambda|\lambda\rangle.$$

Then, by the definition of Hermitian conjugation,

$$\langle\lambda|\mathbf{L}^\dagger = \langle\lambda|\lambda^*.$$

However, since **L** is Hermitian, it is equal to $\mathbf{L}^\dagger$. Thus, we can rewrite the two equations as

$$\mathbf{L}|\lambda\rangle = \lambda|\lambda\rangle \qquad (3.8)$$

and

$$\langle\lambda|\mathbf{L} = \langle\lambda|\lambda^*. \qquad (3.9)$$

Now multiply Eq. 3.8 by <λ| and Eq. 3.9 by |λ>. They become

$$\langle\lambda|\mathbf{L}|\lambda\rangle = \lambda\ \langle\lambda|\lambda\rangle$$

and

$$\langle\lambda|\mathbf{L}|\lambda\rangle = \lambda^*\ \langle\lambda|\lambda\rangle.$$